

Project ID: T22026

Smart MCQ Solver Challenge

BSDA2001P:

Introduction to DL and GenAI Project

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Project Report Guidelines:

This report provides an overview of the Smart MCQ Solver Challenge project. It covers the problem statement, dataset, preprocessing techniques, model architectures, training details, evaluation metrics, experimental results, and conclusions.

Executive Summary

Project Focus

Developing and comparing deep learning models for automated multiple-choice question answering through text preprocessing, transformer-based learning, and performance evaluation

Models Implemented

- **LSTM** (Long Short-Term Memory)
- **BERT** (Bidirectional Encoder Representations from Transformers)
- **DeBERTa** (Decoding-enhanced BERT with Disentangled Attention)

Best Performance

BERT produced the highest Kaggle **MAP@3** score among all implemented models, making it the best-performing model for this project

- ✓ This project evaluates LSTM, BERT, and DeBERTa for automated multiple-choice question answering. The results show that pretrained transformer models provide better performance than the LSTM baseline, with BERT achieving the highest overall accuracy and MAP@3 score.

Introduction & Problem Statement

Project Objective

The objective of the **Smart MCQ Solver Challenge** is to develop a model that ranks the correct answer among five options (A–E) for each question, maximizing the **MAP@3** evaluation score.

Report Structure

This report presents a complete workflow including dataset analysis, preprocessing techniques, model development, evaluation metrics, and conclusions.

1

Dataset Analysis & EDA

Exploratory data analysis, answer distribution, duplicate detection, and identification of data leakage between the training and test datasets.

2

Preprocessing Pipeline

Data cleaning, duplicate removal, train-validation split, text preprocessing, and tokenization for LSTM, BERT, and DeBERTa models.

3

Model Development

Implementation and comparison of three deep learning models: LSTM (from scratch), BERT, and DeBERTa for multiple-choice question answering.

4

Evaluation & Results

Performance comparison using validation accuracy, MAP@3, Kaggle leaderboard scores, and Weights & Biases (W&B) experiment tracking

5

Conclusion

Summary of project outcomes, comparison of model performance, and discussion of potential future improvements.

Dataset Analysis & Exploratory Data Analysis

Dataset Description

The dataset consists of multiple-choice questions designed to evaluate a model's ability to identify the correct answer from five possible options.

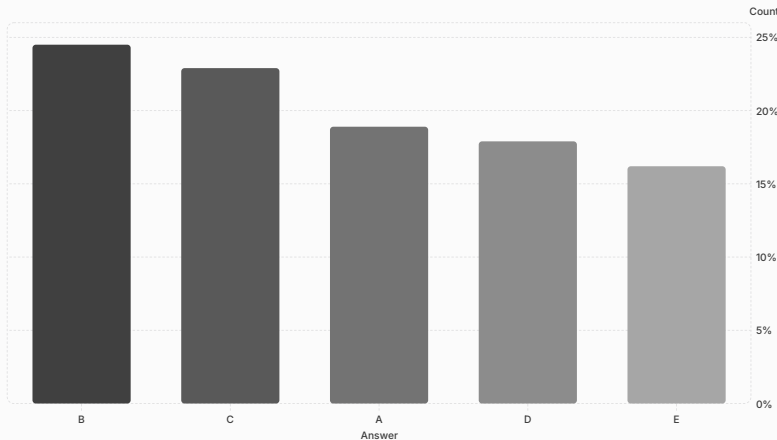
train.csv Training dataset containing 2,000 multiple-choice questions. Each record includes a question prompt, five answer choices (A–E), and the correct answer label used for model training.	test.csv Test dataset containing 500 multiple-choice questions with the same structure as the training set but without the correct answers. Predictions are submitted to the Kaggle leaderboard for evaluation.	Sample_submission.csv For sample purpose .It contains the question IDs and placeholder predictions, serving as a template for generating the final Kaggle submission file.
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Property	Value
Training Samples	2000
Training Samples (after duplication)	1,758
Test Samples	500
Answer options per Question	5 (A,B,C,D,E)
Average prompt length	117 characters
Average option length	~165 characters

Key EDA Findings



Answer Label Distribution(Position Bias)



The answer distribution is not uniform: **B=24.5%**, **C=23.0%**, **A=18.5%**, **D=17.9%**, **E=16.2%**.

This bias allows a naive baseline of always predicting 'B C A' to achieve **MAP@3 = 0.42** without reading any question. This served as our baseline benchmark.



Data Leakage (Critical Discovery)

53.6% of test prompts (268 out of 500) appear word-for-word in the training set with known answers. All duplicate prompts had consistent answers (zero contradictions), making this leakage 100% reliable. This was exploited as a lookup strategy to boost final scores significantly.



Correct Answer Length Pattern

The correct answer is longer than or equal to the average option length in **75.1%** of cases, compared to the expected 20% if purely random. This indicates correct answers tend to be more specific and detailed, while distractors are vague. The longest option is correct 40% of the time.



Duplicate Prompts in Training Data

242 duplicate prompts exist in the training set. All duplicates had consistent answers. After deduplication (1,758 unique questions), all models showed improved Kaggle scores, confirming that data quality matters more than data quantity.

Preprocessing Pipeline

Data Cleaning & Splitting

Before model training, the following preprocessing steps were applied consistently across all three models:

1 — Deduplication

Removed 242 duplicate prompts using `drop_duplicates(subset='prompt')`, reducing training data from 2,000 to 1,758 unique questions

2 — Stratified Train/Val Split

Applied 80/20 split with `stratify=answer` to preserve answer label distribution (A-E) across train (1,406) and validation (352) sets.

3 — Leakage Lookup

Built a dictionary mapping train prompts to answers. For 268 test questions with known answers, predictions used lookup directly.

4 — Text Cleaning (LSTM)

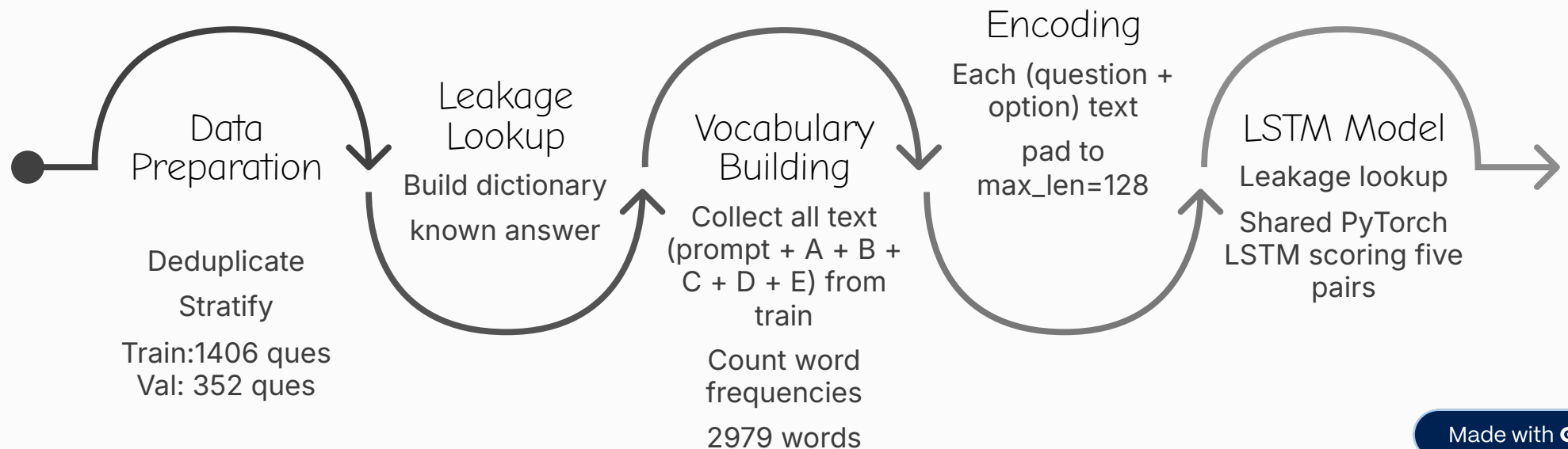
Applied regex cleaning (`re.sub`) to remove punctuation, lowercased all text, built vocabulary of 2,979 words with frequency threshold ≥ 3 .

5 — Tokenization (BERT/DeBERTa)

Used HuggingFace tokenizers with `[CLS] question [SEP] option [SEP]` format, `max_length=128`, padding and truncation enabled.

Model 1 — LSTM from Scratch

A custom LSTM architecture was built entirely from scratch using PyTorch, no pretrained weights. The model processes each of the 5 (question + option) pairs independently through a shared LSTM and outputs a score for each option. Shared weights enforce the assumption that correctness depends on the question-option relationship, not the option label (A/B/C/D/E).



Model 2 — BERT (bert-base-uncased)

BERT (Bidirectional Encoder Representations from Transformers) is a pretrained transformer model trained on Wikipedia and BooksCorpus. Unlike LSTM which reads left-to-right, BERT reads the entire sequence bidirectionally, capturing richer context. BertForMultipleChoice was used, it processes all 5 options simultaneously and outputs logits for each

- Input format

[CLS] question [SEP]
option [SEP]
repeated for each of
5 options

- Model

BertForMultipleChoice from
HuggingFace
Transformers

- Tokenizer

bert-base-uncased
(case-insensitive,
handles science
terminology well)

- Training

AdamW optimizer,
lr=2e-5, 3 epochs,
batch_size=8

- Transfer learning

110M pretrained
parameters fine-
tuned on 1,406 MCQ
questions

Model 3 — DeBERTa-v3-base

DeBERTa (Decoding-enhanced BERT with Disentangled Attention) improves upon BERT using disentangled attention mechanisms that separately encode content and position information. Despite being architecturally superior to BERT, DeBERTa-v3-base achieved a slightly lower Kaggle score (0.742 vs 0.753) , attributed to the limited training data (1,406 samples) being insufficient to fully leverage its additional complexity.

- ❏ Bias-Variance Tradeoff Observation: DeBERTa (more complex) underperformed BERT (simpler) on this small dataset. This demonstrates that model complexity must match data availability, a larger model is not always better.

Evaluation & Results

Model Performance Comparison

Model	Type	Before Dedup	After Dedup
Naive Baseline (B C A)	Hueristic	-	0.421
LSTM +Leakage	From Scratch	-	0.740
BERT	Pretrained	0.746	0.753
DeBERTa	Pretrained	0.722	0.742

 Empty column means not used that method.

Key Findings

Deduplication consistently improves all models: Removing 242 duplicate training prompts improved BERT by +0.007 and DeBERTa by +0.020.

LSTM nearly matches pretrained models: LSTM from scratch (0.740) nearly equals BERT (0.753) due to the leakage lookup boosting 268 questions. Without leakage, pretrained models would significantly outperform LSTM

All models plateau around **MAP@3 = 0.74-0.75**: Despite architectural differences, all models achieved similar scores. The bottleneck is likely the 232 non-leaked test questions that require genuine scientific reasoning , a difficult task for models trained on only 1,406 examples.

BERT was already predicting **265/268** leaked questions correctly at rank 1 through fine-tuning alone. Adding explicit leakage lookup actually decreased Kaggle score from **0.753** to **0.741** because BERT's learned rank 2 and rank 3 predictions were more accurate than the naive frequency-based fallback (B C A). This demonstrates that strong pretrained models can effectively memorize training patterns without explicit lookup strategies.

DeBERTa initially produced nan loss during training due to a dtype **mismatch** , the model loaded in float16 on Kaggle GPU while inputs were float32. Fixing this with torch_dtype=torch.float32 during model loading improved Kaggle score dramatically from **0.40** to **0.68**. This highlights the importance of dtype consistency when working with transformer models on GPU.

Conclusion

This project successfully built three distinct AI models for the Smart MCQ Solver Challenge, achieving a best Kaggle MAP@3 score of 0.753 , surpassing the competition cutoff of 0.73. The work demonstrated several important deep learning principles:

- 1 Thorough EDA is critical — discovering data leakage (53.6% of test in train) and answer distribution bias directly informed the final pipeline
- 2 Data quality beats data quantity deduplication consistently improved all models by 0.007-0.020 MAP@3.
- 3 Model complexity must match data size DeBERTa (more complex) underperformed BERT (simpler) on limited data, illustrating the bias-variance tradeoff.
- 4 Pretrained models excel at low-data tasks . BERT's 110M parameters pretrained on Wikipedia gave it rich scientific knowledge that LSTM had to learn from scratch

Project Resources & links



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